Universitat Rovira i Virgili · Big Data Analytics · Course Project

D1 · Project Proposal

Identifying High-Value Customers, Product Associations and Future Spending in Online Retail Using Apache Spark

Student	**ID**	**Project**	7 · Customer value and baskets
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1. Business problem

The company is a UK-based online retailer selling all-occasion giftware, and many of its customers are wholesalers who reorder regularly. Revenue is likely concentrated in a small share of these customers, but today the company does not know who they are, which of them are drifting away, or which products they tend to buy together. Marketing budget is therefore spread evenly, and promotions are chosen by intuition.

The results will be used by the CRM and marketing manager for three decisions taken each quarter:

- **Who to target:** which customer segments get retention or loyalty actions (e.g. win back valuable customers who have stopped buying) and which need no extra spending.
- **How much to invest per customer:** how to split next quarter's marketing budget according to each customer's *predicted* spending, instead of past spending only.
- **What to offer:** which product pairs to bundle or cross-sell in campaigns and on the website.

2. Analysis, prediction and KPIs

What we will analyze.

1. **Who are the best customers?** For each customer we compute Recency, Frequency and Monetary value (RFM) with Spark, score each dimension in quintiles and group customers into segments such as *Champions*, *Loyal*, *At risk* and *Lost*.
2. **Which products are bought together?** Each invoice is a basket; we compute association rules (support, confidence, lift) with Spark and keep product pairs with high lift and enough support.
3. **Why do valuable customers stop buying?** (our "why?" question) We take customers in the top 20% by spend in a period and compare those who churned (no purchase in the next 90 days) with those who stayed: trend in order frequency and basket size, product mix, cancellation share and country.

What we will predict. Each customer's total spend in the *next quarter* (3 months; 0 if they do not buy). Features use only data before a cutoff date (previous 9 months): RFM, number of orders and distinct products, recent trend, cancellation share and country. The model is a Spark ML gradient-boosted trees regressor. To follow time and the holiday season, we train with cutoff 9 Sep 2010 (target: spend until 9 Dec 2010) and test with cutoff 9 Sep 2011 (target: the last quarter of the data). The **baseline** predicts that each customer spends the same as in the previous quarter; both are compared with MAE and RMSE.

KPIs. Net revenue = Quantity × Price, after removing cancelled invoices; customer KPIs use only rows with a Customer ID.

KPI	Definition	Decision
Revenue concentration	Share of net revenue from the top 20% of customers	Who to target
Repeat-purchase rate	Share of customers active in a quarter who buy again in the next quarter	Who to target
Average order value	Net revenue / number of invoices, by segment and month	How much to invest
Cancellation rate	Share of gross revenue on cancelled invoices (Invoice starting with "C")	Who to target
Basket lift	Lift of the top product pairs: $P(A \cap B)/(P(A)\,P(B))$	What to offer

3. Data and platform

We will use the **Online Retail II** dataset from the UCI Machine Learning Repository.

- **Size and real data:** 1,067,371 rows of real transactions (2009–2011), above the minimum of 1 million rows.
- **License:** Creative Commons Attribution 4.0 International (CC BY 4.0).
- **Link:** `https://archive.ics.uci.edu/dataset/502/online+retail+ii`
- **Platform:** We will build and run the pipeline (data cleaning, Parquet storage and Spark analysis) in **Google Colab** with PySpark.

4. Advanced option and main risks

Advanced option (A · machine learning): We will train a Spark ML gradient-boosted trees regressor to predict each customer's total spend for the next quarter. We will evaluate it with MAE and RMSE.

- **Baseline:** Each customer spends the same next quarter as in the previous quarter.
- **Main risks and mitigation:**
  - *Data leakage:* Using information from after the cutoff date to build the features. We will avoid this with a strict time-based train/test split (September cutoffs).
  - *Skewed spending:* A few very large wholesale buyers can skew the predictions. We will use MAE (less sensitive to outliers) and test a log transformation of the target.
  - *Missing Customer ID:* Over 240,000 rows lack a Customer ID (e.g. guest checkouts) and cannot be used for the customer-level ML model or RFM segmentation. We will mitigate this by filtering these out early and verifying that the remaining 820,000+ identifiable rows are still sufficient to train the gradient-boosted trees effectively.

5. Team roles

Both partners will write, run and be able to explain all the code. Each partner leads these parts:

- **Omar Morales Ortega:** Leads data ingestion, Parquet partitioning, the data-quality log and the association rules analysis (product baskets).
- **Xavier Servin:** Leads the RFM segmentation, the "why?" question and the machine learning prediction pipeline (advanced option).